

Chapter 10 Information System



Introduction

- An information system is a collection of people, procedures, software, hardware, data, and connectivity
 - They all work together to provide information essential to running an organization
 - Computers are used in organizations to keep records of events and to help make decisions
- Competent end users need to understand how the information flows as it moves through an organization

Organizational Information Flow

- Supports the vertically and horizontally flow of information throughout an organization
- View from a functional perspective to examine an organization's structure
 - Can study different functional areas and the different types of people within these functional areas

Functions

Human resources is involved with all human-centered activities across the entire organization. At HealthWise, this department is implementing a new benefits package designed to attract new employees and retain current employees.

Functional
perspective

Accounting tracks all financial activity. At HealthWise, this department records bills and other financial transactions with sporting goods stores. It also produces financial statements, including budgets and forecasts of financial performance.

Research conducts basic research and relates new discoveries to the firm's current or new products department. Research people at HealthWise explore new ideas from exercise physiologists about muscle development. They use this knowledge to design new physical fitness machines.

Marketing handles planning, pricing, promoting, selling, and distributing goods and services to customers. At HealthWise, it even gets involved with creating a customer newsletter that is distributed via the corporate web page.

Production takes in raw materials and people work to turn out finished goods (or services). It may be a manufacturing activity or—in the case of a retail store—an operations activity. At HealthWise, this department purchases steel and aluminum to be used in weight-lifting and exercise machines.



Management Levels

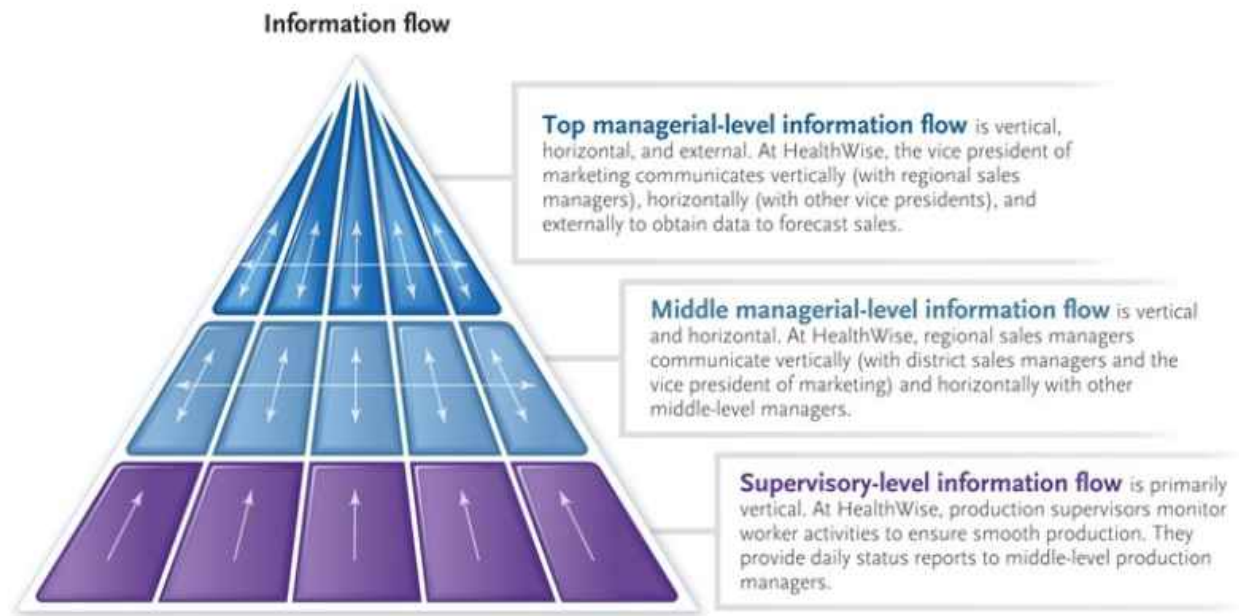


Information Flow Levels

Each level of management has different information needs.

- Require information flows from different levels from different directions
 - Top Managerial-Level Information Flow
 - Vertical, horizontal, and external
 - Middle Managerial-Level Information Flow
 - Vertical and horizontal
 - Supervisory-Level Information Flow
 - Primarily vertical

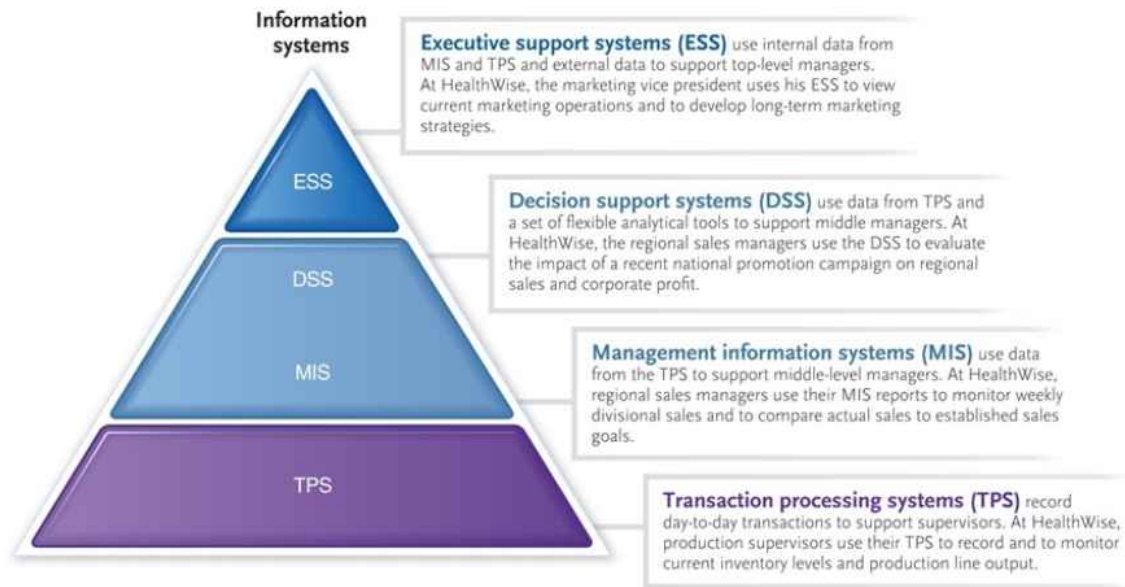
Information Flow Pattern



Summary of Computer-Based Systems

Type	Description
TPS	Tracks routine operations and records events in databases, also known as data processing systems
MIS	Produces standardized reports (periodic, exception, and demand) using databases created by TPS
DSS	Analyzes unanticipated situations using data (internal and external) and decision models (strategic, tactical, and operational)
ESS	Presents summary information in a flexible, easy-to-use, graphical format designed for top executives

Computer-Based Information Systems



Transaction Processing Systems (TPS)

Helps organizations keep track of routine operations and records these events in a database.

- Records day-to-day transactions in a database
- Also called data processing systems (DPS)
- One of the most essential uses of a TPS is in Accounting



TPS for Accounting

- Sales order processing
 - Records the customer requests
- Accounts receivable
 - Records money received from or owed
- Inventory
 - Parts and finished goods in stock
- Purchasing
 - Buying of materials and services
- Accounts Payable
 - Money the company owes
- Payroll
 - Activity involving employer paychecks
- General Ledger
 - Keeps track of all summaries of all the above referenced items

Management Information Systems (MIS)

Computer-based information systems that:

- Produce standardized reports to support decision-making by middle managers
- Integrate data and summarizes details from databases in a structured form
- Produce predetermined reports
 - Periodic reports – produced at regular intervals
 - Exception reports – call attention to unusual events
 - Demand reports – produced on request

HealthWise Group Regional Sales Report			
Region	Actual Sales	Target	Difference
Central	\$166,430	\$175,000	(\$8,570)
Northern	137,228	130,000	7,228
Southern	137,772	135,000	2,772
Eastern	152,289	155,000	(2,711)
Western	167,017	160,000	7,017

Decision Support Systems (DSS)

- Flexible tool for analyzing data for decision-making purposes
- Enables managers to get answers to unexpected and generally non-recurring problems
 - Reports do not have a fixed format
- Consists of four parts:
 - User – anyone who has to make decisions
 - System software – operating system
 - Data - both internal and external
 - Decision models – analytical capabilities
 - 3 basic types

Territory	Month	Sales	Quota
East	May	\$35,000.00	\$44,000.00
East	June	\$40,000.00	\$44,000.00
East	July	\$45,782.00	\$53,998.00
East	August	\$35,982.00	\$39,853.00
Midwest	May	\$23,478.00	\$45,787.00
Midwest	June	\$43,969.00	\$44,764.00
Midwest	July	\$29,488.00	\$49,884.00
Midwest	August	\$23,894.00	\$38,732.00
West	April	\$23,477.00	\$23,757.00
West	June	\$23,478.00	\$23,487.00
West	July	\$23,074.00	\$23,487.00
Midwest	March	\$23,478.00	\$38,748.00
Midwest	May	\$23,847.00	\$23,874.00
Midwest	August	\$23,489.00	\$24,499.00

Decision Models

Give the DSS its analytical capabilities

- Three basic types of decision models
 - Strategic models
 - Assists top level managers in long-range planning
 - Tactical models
 - Assists middle-managers control the work
 - Financial and sales promotion planning
 - Operational models
 - Assists lower-level managers accomplish the daily activities and objectives
- GDSS – Group Decision Support Systems
 - Support multiple or team decision makers

Executive Support Systems (ESS)

- Designed for top management
 - Sophisticated software for presenting, summarizing, and analyzing data, but specifically designed to be easy-to-use
 - Provides immediate access to a company's key performance indicators
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Other Information Systems

- Information workers distribute, communicate, and create information
 - Data workers
 - Knowledge workers
 - Office automation systems (OASs) are designed primarily to support data workers
 - Project managers
 - Videoconferencing systems
 - Knowledge work systems (KWSs) create information in their areas of expertise
 - Used by OAS systems
 - Use specialized systems, such as CAD/CAM
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Expert Systems

Expert Systems aka Knowledge-Based Systems

- Type of artificial intelligence that uses a database to provide assistance to users
 - Knowledge Base is the database holding the data
 - Contains facts and rules to relate these facts distilled from a human expert

Careers In IT

- Information systems managers oversee the work of programmers, computer specialist, systems analysts, and other computer professionals
- Employers look for individuals with strong technical backgrounds, with a Master's degree in business
 - Strong leadership and excellent communications skills
- Information systems managers can expect to earn from \$62,000 to \$100,000 annually



A Look to the Future ~ IBM's Watson

- The Ultimate Information-Finding Machine
 - As information amounts increase, it is more difficult to keep up
 - Can Watson save us?

